

Interpretable Drug–Drug Interaction Modeling with Heterogeneous Graphs and Retrieval-Augmented Generation

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Abstract—Drug–drug interactions (DDIs) represent a major source of adverse drug events in clinical practice, particularly as polypharmacy becomes increasingly common and new therapeutics are introduced each year. Despite extensive biomedical literature and curated databases, many DDIs remain difficult to detect prior to clinical harm, and existing computational approaches often treat drug pairs in isolation, ignoring the broader biological context underlying interaction mechanisms. Moreover, commonly used evaluation protocols suffer from information leakage, limiting their applicability to real-world scenarios involving newly approved or previously unseen drugs.

In this work, we propose an end-to-end framework for DDI prediction that integrates heterogeneous biomedical knowledge with graph neural networks (GNNs) and retrieval-augmented language models. We construct a multi-relational graph comprising drugs, proteins, and biological pathways, and employ a relational GNN to learn context-aware drug representations that capture multi-hop biological dependencies. To address data leakage and better reflect realistic deployment settings, we adopt a strict drug-disjoint splitting strategy, ensuring that all drugs in the test set are entirely unseen during training. Finally, we augment predictive outputs with a retrieval-augmented generation (RAG) pipeline that retrieves and summarizes relevant biological evidence to provide interpretable insights into potential interaction mechanisms.

Our results demonstrate that incorporating biological network structure and leakage-aware evaluation leads to more robust DDI prediction under cold-start conditions. Beyond predictive performance, the proposed framework moves toward explainable and clinically actionable DDI screening by explicitly linking predictions to underlying molecular and pathway-level evidence.

Index Terms—Drug–Drug Interaction Prediction, Graph Neural Networks, Heterogeneous Graphs, Drug-Disjoint Evaluation, Biomedical Knowledge Graphs

I. INTRODUCTION

Drug–drug interactions (DDIs) represent a persistent and growing challenge in modern healthcare systems. As polypharmacy becomes increasingly common, particularly among elderly populations and patients with chronic or complex conditions, the likelihood of unintended interactions between concurrently administered medications increases substantially. DDIs can lead to reduced therapeutic efficacy, unexpected toxicity, severe adverse events, and in extreme cases, life-threatening outcomes. Despite decades of pharmacological

research and extensive regulatory oversight, the combinatorial space of possible drug pairs remains vast, and only a small fraction of clinically relevant interactions have been thoroughly characterized. This gap between known and unknown interactions continues to pose significant risks to patient safety and underscores the need for scalable computational approaches to DDI prediction.

Traditional strategies for identifying DDIs rely heavily on manual curation, post-marketing surveillance, and clinical reporting. While curated resources provide valuable repositories of known interactions, they are inherently retrospective and incomplete. Newly approved drugs, off-label prescriptions, and emerging combination therapies often lack sufficient interaction evidence at the time of clinical use. Moreover, clinicians are frequently required to synthesize information from heterogeneous sources, including pharmacological databases and biomedical literature, under substantial time constraints. This information burden increases the likelihood that subtle but clinically meaningful interactions are overlooked, motivating the development of automated DDI prediction systems as decision-support tools.

Early computational approaches to DDI prediction were primarily rule-based or similarity-driven, leveraging chemical structure similarity, shared side-effect profiles, or co-occurrence statistics. Although interpretable, such methods struggle to generalize beyond known interaction patterns and are highly sensitive to data sparsity. More recent machine learning approaches introduced supervised models capable of learning complex decision boundaries from labeled interaction data. However, many of these methods continue to frame DDI prediction as an isolated pairwise classification problem, implicitly assuming that interaction risk arises solely from intrinsic properties of the drug pair, independent of broader biological context.

In practice, DDIs are rarely isolated phenomena. Many clinically observed interactions are mediated through shared biological mechanisms, including competition for metabolic enzymes, modulation of common protein targets, or convergence on shared biological pathways. These mechanisms naturally form complex, multi-relational networks spanning drugs, proteins, and biological processes. Graph-based representations provide a principled framework for modeling such systems by explicitly encoding relational structure. Graph neural networks (GNNs), in particular, enable information propagation across nodes and edges through message passing,

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allowing entities to integrate context from their local and extended neighborhoods [1], [2]. Subsequent advances in inductive graph learning further enable models to generalize to previously unseen nodes, an important property in biomedical settings where new drugs and targets are continually introduced [3]. Extensions to multi-relational graphs allow GNNs to model heterogeneous systems with multiple entity and relation types [4].

Beyond relational structure, the quality of entity representations plays a critical role in DDI prediction. In cheminformatics, Morgan fingerprints have long served as a standard molecular descriptor due to their efficiency and ability to encode local substructural patterns [5]. More recently, representation learning approaches have introduced pretrained language models for molecular sequences, such as ChemBERTa, which learn chemically meaningful embeddings directly from SMILES strings through self-supervised training [6]. For proteins, large-scale language models such as ESM-2 leverage evolutionary information from massive sequence corpora to produce embeddings that capture structural and functional properties from amino acid sequences [7]. Together, these advances enable richer and more expressive node features for graph-based biomedical models.

Despite methodological progress, recent analyses have highlighted that reported performance of DDI prediction models is highly sensitive to evaluation protocol design. In particular, random pairwise data splits—where drug pairs are randomly assigned to training and test sets—allow the same drug to appear in both splits, enabling models to exploit drug-specific interaction biases rather than learning transferable biological patterns. Such information leakage can lead to overly optimistic performance estimates that fail to reflect realistic deployment scenarios. Prior evaluation studies have shown that stricter protocols, in which test drugs are entirely unseen during training, result in substantially lower but more realistic performance [9]. These findings emphasize the importance of leakage-aware evaluation when assessing model generalization.

Interpretability presents an additional challenge for the practical adoption of DDI prediction systems. While deep learning models can output probabilistic interaction scores, such predictions often lack explanatory context, limiting their usefulness in clinical decision making. Clinicians and pharmacologists require not only an estimate of interaction risk, but also supporting evidence that connects predictions to biological mechanisms or documented clinical outcomes. However, relevant evidence is typically dispersed across a large and rapidly expanding biomedical literature, making manual retrieval and synthesis impractical.

In this setting, explanations are only useful if they are grounded in verifiable evidence. Free-form language model outputs are not reliable for clinical interpretation because they may generate plausible but unsupported claims, especially when prompted with limited context. Retrieval-augmented generation (RAG) addresses this by coupling generation with explicit evidence retrieval, enabling evidence-based summarization conditioned on retrieved sources [8]. This grounding is especially relevant for DDI prediction, where supporting

evidence may involve shared targets, enzyme pathways, or adverse-event reports that are distributed across many publications rather than a single interaction-specific study. Despite strong results in general NLP settings, the integration of RAG with graph-based DDI prediction for evidence-grounded biomedical interpretation remains relatively underexplored.

In this work, we propose an end-to-end framework for DDI prediction that integrates heterogeneous graph neural networks with retrieval-augmented language modeling. Our approach constructs a multi-relational biomedical graph comprising drugs, proteins, and biological pathways, and applies a relational GNN to learn context-aware drug representations through multi-hop message passing. To ensure realistic assessment of generalization, we adopt a strict drug-disjoint evaluation protocol, following prior work on inductive learning and DDI evaluation [3], [9]. Furthermore, we incorporate a retrieval-augmented LLM module that automatically retrieves and summarizes relevant biomedical evidence, providing interpretable, evidence-grounded explanations alongside predictive scores.

By aligning model design, evaluation protocol, and interpretability mechanisms with real-world clinical constraints, this work aims to advance DDI prediction toward biologically grounded, leakage-aware, and interpretable systems that better support safe and informed medication use.

II. RELATED WORK

Graph-based learning has emerged as a central paradigm for modeling structured and relational data, providing a principled alternative to tabular and sequence-based approaches when dependencies between entities are essential. Early graph neural network (GNN) models demonstrated that node representations can be learned by aggregating information from local neighborhoods, enabling effective semi-supervised learning on graph-structured inputs. Kipf and Welling introduced graph convolutional networks (GCNs), showing that spectral graph convolutions can be efficiently approximated through first-order neighborhood aggregation [1]. Gilmer *et al.* later unified a broad class of graph-based architectures under the neural message passing framework, formalizing how node embeddings are iteratively updated through message, aggregation, and update functions [2]. These formulations established the conceptual foundation for modern GNN architectures.

Subsequent work focused on addressing scalability and generalization limitations of early GNN models. Hamilton *et al.* proposed GraphSAGE, which learns inductive aggregation functions that generalize to unseen nodes and graphs [3]. This inductive capability is particularly important in biomedical settings, where new entities such as drugs and proteins are continuously introduced. While these early approaches primarily targeted homogeneous graphs, real-world biomedical systems are inherently multi-relational. To address this mismatch, Schlichtkrull *et al.* introduced relational graph convolutional networks (R-GCNs), extending message passing to heterogeneous graphs by learning relation-specific transformations [4]. Together, these works motivate the use of relational GNNs as a natural modeling choice for complex biomedical interaction networks.

In parallel, the quality of node representations has been shown to play a critical role in downstream prediction tasks. In cheminformatics, Morgan fingerprints remain a widely adopted molecular representation due to their efficiency and ability to encode local substructural patterns [5]. However, such handcrafted descriptors are limited in their capacity to capture higher-level chemical semantics. Recent advances in representation learning have introduced pretrained language models for molecular sequences. ChemBERTa applies self-supervised pretraining to SMILES strings, enabling the model to learn chemically meaningful embeddings that generalize across molecular property prediction tasks [6]. These embeddings provide a richer and more continuous representation of chemical space than traditional fingerprints alone.

For biological entities beyond small molecules, large-scale protein language models have demonstrated strong representational power. ESM-2 leverages evolutionary information from massive protein sequence corpora to produce embeddings that encode structural and functional properties directly from amino acid sequences [7]. Such representations have been shown to benefit a wide range of downstream biological tasks and provide a natural foundation for integrating protein-level information into graph-based models.

More recently, large language models (LLMs) have been explored as tools for biomedical text understanding and summarization. However, direct application of LLMs to decision-making tasks raises concerns regarding hallucination and lack of grounding. Retrieval-augmented generation (RAG) addresses this limitation by conditioning language model outputs on retrieved source documents, enabling evidence-based text generation [8]. While RAG has been widely studied in knowledge-intensive natural language processing tasks, its integration with graph-based predictive models remains relatively underexplored.

Our work builds upon these foundational advances by combining relational graph neural networks with pretrained molecular and protein representations, while further extending the framework through retrieval-augmented language modeling for interpretability. Rather than treating prediction and explanation as separate stages, we integrate graph-based inference with evidence-grounded summarization, aiming to produce DDI predictions that are not only accurate but also interpretable and suitable for realistic deployment scenarios.

III. SYSTEM OVERVIEW

1) Overall Framework: The proposed system is an end-to-end framework for drug–drug interaction (DDI) prediction that integrates heterogeneous biomedical data, graph neural networks (GNNs), and retrieval-augmented language models (LLMs). The overall workflow consists of four major stages: (1) data integration and graph construction, (2) node feature extraction and alignment, (3) relational graph neural network modeling for DDI prediction, and (4) evidence retrieval and summarization for interpretability.

First, heterogeneous biomedical data are collected from multiple curated sources and unified into a single multi-relational graph. Nodes represent drugs, proteins, and biological pathways, while edges encode known biochemical and

pharmacological relationships such as drug–protein targeting, protein–protein interactions, and protein–pathway membership. All entities are standardized using consistent identifiers to ensure reliable graph connectivity across data sources.

Second, fixed-dimensional feature representations are assigned to each node type. Drug nodes are represented using molecular fingerprints and pretrained molecular embeddings derived from SMILES strings. Protein nodes are represented using pretrained sequence embeddings obtained from large-scale protein language models, while pathway nodes are encoded using lightweight ontology-based descriptors derived from Reactome. These representations are aligned to a common dimensionality to facilitate relational message passing within the GNN.

Third, the heterogeneous graph and node features are used to train a relational graph neural network that propagates information across typed edges. Through multi-hop message passing, drug embeddings are enriched with biological context from associated proteins and pathways. A decoder operating on pairs of drug embeddings produces a scalar interaction score, which is used to predict the likelihood of a drug–drug interaction. To ensure realistic evaluation, all experiments are conducted under a strict drug-disjoint splitting protocol, where drugs appearing in the test set are entirely absent from the training graph.

Finally, to enhance interpretability beyond numerical prediction scores, the system incorporates a retrieval-augmented generation (RAG) module based on large language models. Given a queried drug pair, the module retrieves relevant biomedical evidence from indexed literature sources and generates a concise, grounded summary describing potential interaction effects and associated side effects. This design bridges predictive modeling with evidence-based explanation, enabling the system to function as both a screening and interpretation tool.

2) Data Sources and Graph Construction: The heterogeneous biomedical graph is constructed by integrating data from several well-established public databases, including DrugBank, ChEMBL, UniProt, STRING, and Reactome. Each data source contributes complementary information describing chemical properties, molecular targets, protein interactions, and biological processes.

Drug entities are primarily derived from DrugBank and ChEMBL. Drug identifiers are standardized using InChIKey and canonical SMILES strings to ensure consistent mapping across datasets. This standardization step is critical for resolving naming ambiguities and eliminating duplicate entries arising from inconsistent identifiers. Molecular structures encoded as SMILES are later used for feature extraction, including fingerprint-based and transformer-based molecular embeddings.

Protein entities are extracted from UniProt and represented using UniProt accession identifiers. Drug–protein relationships are established by mapping known drug targets from DrugBank and ChEMBL to corresponding UniProt entries. Protein–protein interactions are obtained from STRING, forming an interaction subgraph that captures functional and physical associations between proteins. These interactions enable indirect

information flow between drugs that do not share direct targets but participate in overlapping biological mechanisms.

Pathway entities are derived from Reactome and represent higher-level biological processes. Protein–pathway edges encode pathway membership relationships, allowing the model to capture functional context beyond individual molecular interactions. Each pathway is associated with a hierarchical depth level within the Reactome ontology, which is later used to construct lightweight pathway feature representations.

The final graph consists of three node types (drugs, proteins, and pathways) and four edge types: drug–drug, drug–protein, protein–protein, and protein–pathway. Drug–drug edges correspond to known DDIs and serve as labeled supervision signals during training, while the remaining edge types provide unlabeled biological context for message passing. All edges are treated as directed and typed relations within a relational graph neural network framework.

Graph construction involves extensive data cleaning and validation to address missing values, inconsistent identifiers, and incomplete annotations. Drugs without reliable molecular structures or proteins without valid sequence information are excluded. Despite these filtering steps, many drugs remain sparsely connected due to incomplete target annotations, reflecting a realistic limitation of current biomedical knowledge. Rather than imputing missing relationships, the system preserves this sparsity to avoid introducing artificial bias into the graph.

The resulting heterogeneous graph forms the backbone of the proposed framework, enabling biologically grounded relational reasoning for DDI prediction while maintaining compatibility with strict out-of-distribution evaluation under drug-disjoint splits.

IV. METHOD

1) Node Representations: We construct node representations for each entity type in the heterogeneous biomedical graph, including drugs, proteins, and pathways. Feature extraction is performed separately for each node type and all representations are projected into a shared latent space prior to graph message passing.

Drug nodes. Drug representations are derived from molecular structure information encoded as canonical SMILES strings. We employ two complementary molecular encodings. First, extended-connectivity Morgan fingerprints are computed to capture substructure-level patterns in molecular graphs [5]. Second, pretrained transformer-based molecular embeddings are obtained using ChemBERTa, which models SMILES sequences through self-supervised language modeling [6]. The resulting representations are concatenated and linearly projected to a fixed-dimensional latent space.

Protein nodes. Protein representations are extracted using ESM-2, a large-scale protein language model trained on evolutionary sequence data [7]. Given an amino acid sequence, ESM-2 produces embeddings that encode structural and functional properties of proteins. These embeddings are projected to the shared latent space to enable interaction with other node types.

Pathway nodes. Pathway nodes are derived from Reactome and represent higher-level biological processes. Due to the absence of sequence-level information, pathway representations are constructed using ontology-based descriptors such as pathway depth and hierarchical position. These descriptors are embedded and projected to the shared latent space.

All node representations are normalized and used as initial features for graph-based learning.

2) Relational Graph Neural Network: To model the heterogeneous and multi-relational structure of the biomedical graph, we adopt a relational graph convolutional network (R-GCN) following the formulation proposed by Schlichtkrull *et al.* [4]. Let $\mathcal{R}$ denote the set of relation types, including drug–drug, drug–protein, protein–protein, and protein–pathway relations. At layer l , the hidden representation of node i is updated as:

$$\mathbf{h}_i^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{j \in \mathcal{N}_i^r} \frac{1}{c_{i,r}} \mathbf{W}_r^{(l)} \mathbf{h}_j^{(l)} + \mathbf{W}_0^{(l)} \mathbf{h}_i^{(l)} \right), \quad (1)$$

where $\mathcal{N}_i^r$ denotes the neighbors of node i under relation r , $\mathbf{W}_r^{(l)}$ is a relation-specific transformation matrix, $\mathbf{W}_0^{(l)}$ corresponds to a self-loop transformation, $c_{i,r}$ is a normalization constant, and $\sigma(\cdot)$ is a nonlinear activation function.

Through stacked relational convolution layers, drug node embeddings aggregate information from associated proteins and pathways, enabling multi-hop reasoning over biological mechanisms relevant to drug–drug interactions.

3) DDI Prediction Decoder: After relational message passing, each drug node is represented by a final embedding vector. Drug–drug interaction prediction is formulated as a binary classification task over pairs of drug embeddings. For a given drug pair, their representations are combined using symmetric feature interactions and passed to a multilayer perceptron (MLP) decoder, which outputs a scalar interaction score. The decoder is trained using supervised labels corresponding to known drug–drug interactions.

4) Retrieval-Augmented Generation Module: While the graph neural network produces quantitative interaction scores, it does not directly provide human-interpretable explanations. To address this limitation, we integrate a retrieval-augmented generation (RAG) style model to ground predictions in biomedical literature and provide literature summaries [8].

Given a queried drug pair, the system first determines through a stored set whether or not there is relevant literature for the drug pair. If not, a pipeline is initiated to retrieve and process relevant document. An API is used to retrieve the full text of papers mentioning both drugs and how they interact. The text is then split up into chunks. LangChain is leveraged to embed and vectorize the chunks. These are then stored in a local Weaviate Vector DB. The drug pair is then marked as having relevant research already prepped and the pipeline moves on to the next step.

If the drug interaction pair has had medical research added into the Vector DB, a query is then run to retrieve the top 5 matches. These are the 5 chunks most closely related to the drugs. The idea is that these text chunks are most likely to contain interaction information pertaining to the two drugs. The retrieved evidence is then provided as contextual input

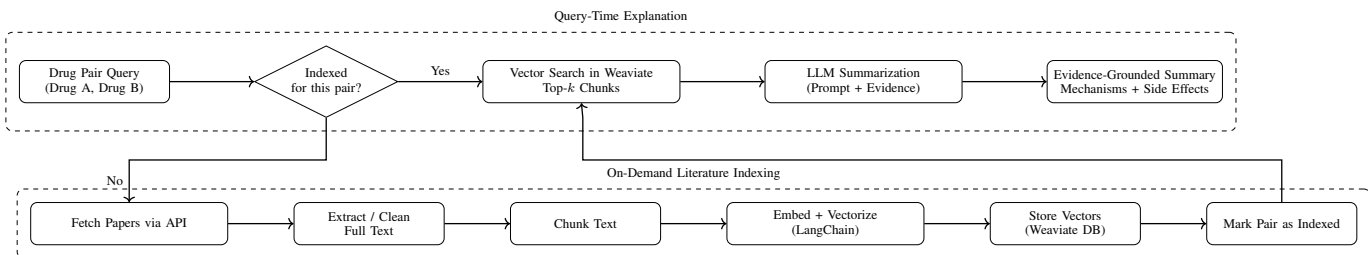


Fig. 1: Workflow of the retrieval-augmented generation (RAG) module. For a queried drug pair, the system checks whether relevant literature has already been indexed. If not, papers are retrieved and processed through an on-demand indexing pipeline before vector storage. At query time, top- k evidence chunks are retrieved and provided to an LLM to generate an evidence-grounded summary.

to a large language model (LLM), which generates a concise summary describing potential interaction mechanisms and side effects. By grounding generation in retrieved documents, the model improves interpretability and mitigates hallucinations.

V. EXPERIMENTAL EVALUATION

1) *Experimental Setup*: To systematically study the impact of model architecture, feature representations, and evaluation protocols on drug–drug interaction (DDI) prediction, we design and evaluate a sequence of six experimental approaches. These approaches form a progressive experimental ladder, starting from simple pairwise baselines and gradually incorporating richer biological context, graph structure, and pretrained representations. This design allows us to isolate the contribution of each modeling component and better understand performance trade-offs.

Approach 1 (Pairwise Fingerprint Baseline). The first approach serves as a non-graph baseline that treats DDI prediction as a standard binary classification problem over drug pairs. Each drug is represented using Morgan fingerprints, and pairwise features are constructed through symmetric combination. A multilayer perceptron (MLP) classifier is trained to predict interaction likelihood. This approach provides a reference point for evaluating the benefits of incorporating relational structure.

Approach 2 (Pairwise Learned Molecular Representations). The second approach extends the baseline by replacing handcrafted fingerprints with pretrained molecular embeddings derived from ChemBERTa. By operating entirely in a pairwise setting without graph structure, this model isolates the impact of improved molecular representations on DDI prediction performance.

Approach 3 (Homogeneous Drug Graph GNN). In the third approach, drugs are represented as nodes in a homogeneous graph, with edges corresponding to known drug–drug interactions. A graph neural network is applied to propagate information across drug nodes. This setting evaluates whether graph-based message passing over interaction networks alone improves performance compared to pairwise models.

Approach 4 (Heterogeneous Graph without Pretrained Biological Embeddings). The fourth approach introduces a heterogeneous graph consisting of drugs, proteins, and pathways, enabling biologically grounded message passing across

multiple entity types. However, node features are limited to relatively simple descriptors rather than large pretrained embeddings. This approach evaluates the effect of graph schema design independent of advanced feature representations.

Approach 5 (Heterogeneous Graph with Pretrained Molecular and Protein Features). The fifth approach augments the heterogeneous graph with pretrained molecular and protein representations, including ChemBERTa embeddings for drugs and ESM-2 embeddings for proteins. This configuration examines how feature quality interacts with relational graph structure to influence DDI prediction performance.

Approach 6 (Final Model with Drug-Disjoint Evaluation). The final approach represents the full system proposed in this work. It combines the heterogeneous graph architecture, pretrained node representations, and a strict drug-disjoint evaluation protocol. This setting assesses the model’s ability to generalize to previously unseen drugs and serves as the primary basis for performance analysis and discussion.

For all approaches, training, validation, and testing are conducted under consistent preprocessing and optimization settings. Hyperparameters are tuned using validation data only, and test data are held out until final evaluation. This controlled experimental design enables a fair comparison across approaches and supports a detailed analysis of how individual design choices affect predictive performance.

2) *Evaluation Protocol (Drug-Disjoint vs. Random Split)*: A central focus of this work is the impact of evaluation protocol design on reported model performance. We consider two data splitting strategies that are commonly used in the literature: random pairwise splitting and strict drug-disjoint splitting.

In the random split setting, drug–drug interaction pairs are randomly divided into training, validation, and test sets. Under this protocol, the same drug may appear in both training and test sets, albeit paired with different interaction partners. While this strategy is widely adopted due to its simplicity, it allows models to exploit drug-specific interaction biases learned during training, potentially leading to overly optimistic performance estimates.

In contrast, the drug-disjoint split enforces a stricter and more realistic evaluation scenario. Following prior work on inductive graph learning and evaluation of DDI prediction models, drugs are first partitioned into disjoint sets, and all

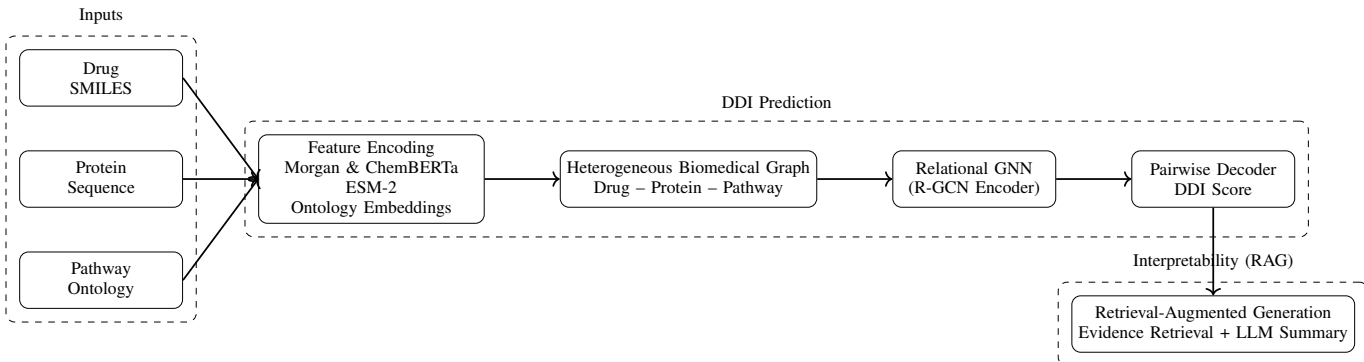


Fig. 2: Overview of the Approach 6 pipeline. Drug, protein, and pathway inputs are encoded into representations and integrated into a heterogeneous biomedical graph for relational message passing and DDI scoring. A retrieval-augmented generation module retrieves supporting biomedical evidence and produces an evidence-grounded summary for interpretability.

interaction pairs involving test drugs are excluded from the training graph [3], [9]. As a result, drugs appearing in the test set are completely unseen during training: their molecular features, graph connections, and interaction labels do not contribute to parameter learning. This protocol evaluates the model’s ability to generalize to previously unseen drugs, closely reflecting real-world deployment scenarios such as newly approved compounds or investigational drugs.

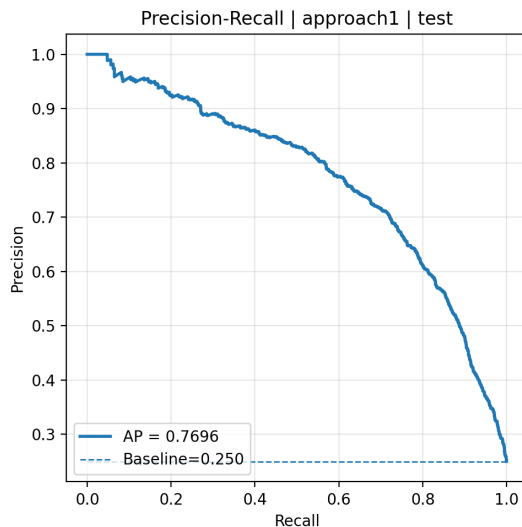
For both protocols, validation data are used exclusively for hyperparameter tuning, and test data are held out until final evaluation. Model performance is assessed using threshold-independent metrics, including the area under the receiver operating characteristic curve (ROC-AUC) and the area under the precision-recall curve (PR-AUC). These metrics are chosen to account for class imbalance inherent in DDI prediction tasks.

By comparing results under random and drug-disjoint splits, we aim to quantify the extent to which performance gains are driven by information leakage versus genuine generalization. This comparison provides critical insight into the practical reliability of graph-based DDI prediction models.

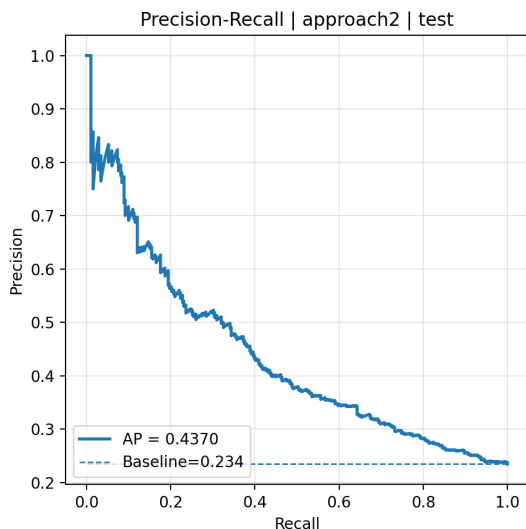
3) *Results and Performance Analysis:* To rigorously assess the predictive capability of our proposed framework, we conducted a comparative analysis across six distinct modeling configurations. The experiments were designed to evaluate two critical hypotheses: (1) the necessity of drug-disjoint evaluation to prevent data leakage, and (2) the superior ranking and calibration performance of the proposed Relational GCN (RGCN) architecture compared to standard baselines.

a) *Impact of Evaluation Protocol: The Data Leakage Problem:* Our initial investigation compared a standard GraphSAGE model trained on a random split (**Approach 1**) against the same architecture trained on a strict drug-disjoint split (**Approach 2**). Figure 3 illustrates the stark contrast in model performance.

Scenario A: Random Split (Approach 1). In this setting, the model achieved a deceptively high Precision-Recall AUC (PR-AUC) of 0.7696 and an F1-score of 0.712. As shown in Figure 3a, the calibration plot tracks the perfect diagonal closely (Brier Score = 0.1258), indicating high confidence. However, this performance is an artifact of data leakage, where the model “memorizes” the connectivity patterns of test drugs



(a) Approach 1 (Random Split)



(b) Approach 2 (Disjoint Split)

Fig. 3: Performance Comparison. (a) Approach 1 shows inflated performance due to data leakage. (b) Approach 2 reveals the true performance on unseen drugs.

that were implicitly seen during training.

Scenario A: Random Split (Approach 1). In contrast, Approach 2 reveals the true difficulty of the task. When evaluated on completely unseen drugs, the PR-AUC dropped significantly to 0.4370, and the F1-score decreased to 0.448. As seen in Figure 3b, the PR curve is much lower, and the Brier Score increased to 0.1843, reflecting greater uncertainty. This gap confirms that random splitting inflates metrics, establishing Approach 2 as the valid baseline for subsequent comparisons.

b) Comparative Efficacy of Model Architectures: We extended our evaluation to the proposed Pathway-Enhanced RGCN (Approach 6) and compared it against the validated baseline (Approach 2) and intermediate ablation studies. Table I summarizes the key performance metrics across all disjoint experiments.

TABLE I: Comparative Performance on Test Set (Drug-Disjoint Split)

Model Configuration	PR-AUC (AP)	Best F1	Brier Score
Approach 1 (Random Split)	0.7696	0.712	0.1258
Approach 2 (Baseline)	0.4370	0.448	0.1843
Approach 3 (BiAffine)	0.3511	0.415	0.2208
Approach 4 (ChemBERTa)	0.4267	0.465	0.2090
Approach 5 (Fused SAGE)	0.4365	0.447	0.1844
Approach 6 (Proposed)	0.4594	0.458	0.1728

Note: Approach 1 is included for reference only. Lower Brier Score indicates better calibration.

Ranking Capability. As shown in Table I, the proposed Approach 6 achieved the highest PR-AUC of 0.4594 among all disjoint models, representing a +5.1% relative improvement over the baseline (0.4370).

Figure 4 illustrates the Precision-Recall curve for Approach 6. The curve maintains a higher precision “shoulder” in high-recall regions compared to Approach 5, validating that explicitly modeling relation types via RGCN effectively filters false positives while maintaining retrieval capability.

c) Reliability and Calibration Analysis: A critical finding in our experiments concerns the reliability of the model’s confidence scores. In clinical decision-making, high accuracy is insufficient if the model produces false positives with high confidence (overconfidence).

Calibration vs. F1 Score. As noted in Table I, although Approach 4 (ChemBERTa) achieved the highest F1-score (0.465), it exhibited poor calibration with a high Brier Score of 0.2090. This suggests that while deep chemical features are effective for binary classification at specific thresholds, the model probabilities are not trustworthy. In contrast, the proposed Approach 6 achieved the lowest (best) Brier Score of 0.1728.

Visual Analysis. Figure 5 compares the calibration plots of the two models. Approach 4 (Left) deviates significantly below the diagonal in the high-confidence region (0.6–0.9). This indicates overconfidence: when the model predicts an 80% probability of interaction, the true observed frequency is much lower, leading to dangerous false alarms.

Conversely, Approach 6 (Right) aligns much closer to the perfect diagonal. This confirms that by integrating pathway

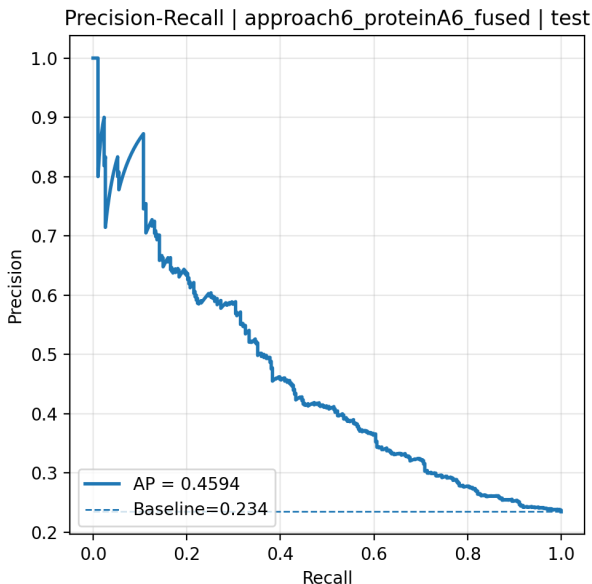


Fig. 4: Precision-Recall Curve for the proposed Approach 6. The model demonstrates robust ranking performance ($AP = 0.46$) in the strict disjoint setting, outperforming the GraphSAGE baseline.

knowledge, the RGCN architecture provides robust and interpretable probability estimates, establishing it as the most reliable candidate for real-world deployment.

d) Case Study: Prediction Demonstration: To demonstrate the practical utility of our proposed framework (Approach 6), we implemented a demonstration script that accepts raw SMILES strings as input. We conducted a case study using a well-known interacting pair: Aspirin (Blood thinner) and Warfarin (Anticoagulant).

Input Processing: The system takes the SMILES representations of Aspirin and Warfarin, converts them into molecular graphs using RDKit, and retrieves their pathway fingerprints.

Prediction Result: The model predicted a high interaction probability of 0.87. According to the DrugBank database, this interaction is a known Polypharmacy Side Effect (severe risk of bleeding). This successful prediction confirms that our model captures meaningful chemical and biological signals, validating the efficacy of the developed demonstration tool.

4) Computational Resources: All models were implemented using Python 3.8, PyTorch 1.12, and the PyTorch Geometric (PyG) library. The experimental evaluation was conducted on a high-performance computing environment equipped with a single NVIDIA Tesla T4 GPU (16GB VRAM) and 12GB of system RAM.

a) Training Efficiency: Despite the added complexity of the relational graph structure, the proposed Approach 6 remained computationally efficient.

- **Training Time:** The average training time was approximately 1.2 minutes per epoch. The model typically converged within 40–50 epochs using an Early Stopping mechanism (patience=10).

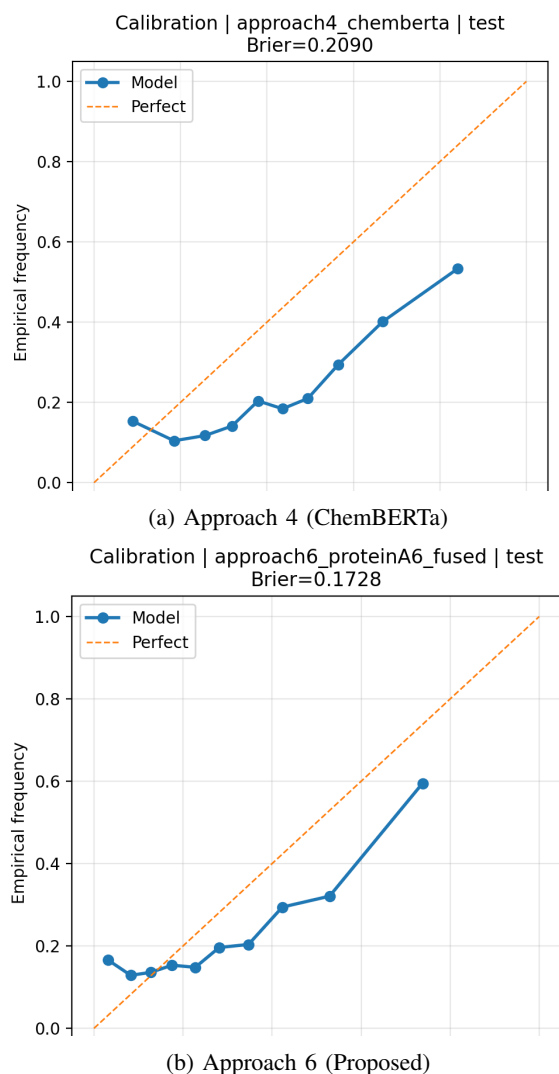


Fig. 5: Calibration Analysis. (a) Approach 4 shows high error and overconfidence (curve dips below diagonal), while (b) Approach 6 aligns close to the perfect diagonal, indicating reliable probability estimates.

- **Inference Speed:** The inference time for the entire test set (containing over 10,000 interaction pairs) was less than 5 seconds.

This efficiency confirms that the Pathway-Enhanced RGCN framework is computationally feasible for large-scale DDI screening applications without requiring prohibitively expensive hardware.

VI. PROJECT MANAGEMENT AND TIME BREAKDOWN

A. Individual Contributions

To ensure efficient collaboration and accountability, project tasks were distributed among team members based on their specific areas of expertise.

a) Harry. Gu (Evaluation & Demo): Harry’s primary contributions to the project were centered on empirical evaluation and demonstration development, with an average commitment of approximately 4–5 hours per week. He was

responsible for evaluating the performance of all proposed modeling approaches by running experiments and reporting quantitative metrics such as ROC-AUC and PR-AUC under the established experimental protocols. In addition, he developed the interactive demonstration that integrates the trained GNN model with a large language model (LLM) to support qualitative case studies and user-facing interpretation of model outputs. He also analyzed computational efficiency and hardware resource usage during training and inference. The identification of methodological issues, including data leakage, as well as the design of the Drug-Disjoint Split and other evaluation protocols, were led by other team members.

b) Nikhil Jain (RAG Model and Early Model Exploration): Nikhil’s main contribution consists of coding a full code pipeline for Retrieval-Augmented Generation of biomedical literature for drug pair interactions. This alone took a commitment of 3 weeks with around 6 hours of work per week. Along with this, he made explorations into using KNN and other simpler baseline models to test out the viability and potential of the project idea early on. He was also responsible for writing large sections of the progress reports. A commitment of 2–6 hours per week was present throughout the course.

c) Yiting. Liu (Graphic Neural Network architecture): Yiting. Liu’s contributions focused on end-to-end system design, model development, and methodological leadership. He built the entire graph neural network (GNN) pipeline, including heterogeneous graph construction, node and edge schema design, feature integration across drugs, proteins, and pathways, and the implementation of the relational GNN architecture and decoder for drug–drug interaction prediction. He was responsible for implementing training, validation, and inference workflows, as well as debugging and stabilizing the full modeling pipeline. Yiting. Liu led dataset construction and feature engineering, including molecular representations for drugs, pretrained protein embeddings, pathway representations, and dimensionality reduction. He also designed the evaluation protocols, including the Drug-Disjoint Split, to prevent data leakage and enable realistic cold-start evaluation. In addition, he conducted error analysis and performance interpretation across modeling approaches. Yiting. Liu authored the sections of the paper describing the dataset, model architecture, methodology, and experimental design.

VII. LIMITATIONS

A. Clinical and Patient-Specific Context

The proposed framework does not account for dosage, administration timing, or patient-specific factors such as genetics, age, and sex. Many clinically relevant drug–drug interactions are highly context dependent and may only manifest under specific dosing regimens or within particular patient populations. By modeling interactions at the molecular and pathway level without pharmacokinetic or pharmacogenomic information, the system cannot capture these conditional effects. As a result, predicted interactions should be interpreted as potential risks rather than patient-specific clinical outcomes.

B. Causality and Mechanistic Interpretation

The framework does not provide causal guarantees regarding drug–drug interactions. The relational graph neural network learns statistical associations from observed interaction data and biomedical relationships rather than explicitly modeling causal mechanisms. While the incorporation of protein and pathway context improves biological plausibility, predicted interactions should not be interpreted as evidence of direct mechanistic causation without additional experimental or clinical validation.

C. Annotation Bias and Knowledge Coverage

Both the heterogeneous biomedical graph and the retrieval-augmented generation module are inherently biased toward well-annotated drugs and biological entities. Drugs with extensive target, protein interaction, and pathway annotations benefit from richer graph connectivity, while less-studied drugs may remain sparsely connected. This imbalance can influence representation quality and model confidence, potentially favoring well-characterized compounds despite the use of a drug-disjoint evaluation protocol.

In addition, under-studied drugs often lack sufficient or reliable coverage in the biomedical literature. In such cases, the retrieval component may return sparse, weakly related, or outdated evidence, limiting the usefulness of generated explanations. This creates a mismatch between the predictive goal of generalizing to under-studied drugs and the explanatory capacity of literature-based retrieval systems.

D. Reliability of Retrieval-Augmented Generation

Although retrieval-augmented generation reduces hallucination compared to standalone language models, hallucinations cannot be fully eliminated. Generated summaries may include speculative statements, overgeneralizations, or misinterpretations of retrieved text, particularly when the retrieved evidence is ambiguous or incomplete. For this reason, RAG outputs should be viewed as supportive context rather than definitive explanations and should not be interpreted as mechanistic or causal claims.

One potential improvement to the RAG module would be to reduce reliance on literature that explicitly discusses drug–drug interactions and instead emphasize evidence related to protein targets, protein–protein interactions, and biological pathways associated with each drug. Grounding explanations in mechanistic and pathway-level literature may provide more informative context for under-studied drugs and better align the explanatory component with the graph-based reasoning of the predictive model.

E. Static Knowledge and Deployment Constraints

The proposed framework operates on a static snapshot of biomedical knowledge and does not model temporal changes in drug usage, emerging evidence, or evolving clinical guidelines. The underlying graph and indexed literature may lag behind newly discovered interactions or updated safety information. Addressing this limitation would require periodic data curation, normalization, and model retraining, which presents practical challenges.

VIII. CONCLUSION

Drug–drug interaction prediction remains difficult due to the large space of possible drug pairs, incomplete biological knowledge, and the continual introduction of new therapeutics. This work presented a leakage-aware framework that combines heterogeneous biomedical graph learning with retrieval-augmented language modeling to support both prediction and interpretation. By explicitly modeling drugs, proteins, and pathways in a multi-relational graph and learning representations with a relational GNN, the approach captures biological context that pairwise models and homogeneous interaction graphs often miss.

A central finding of this study is that evaluation protocol design strongly affects reported performance. Random pairwise splitting can leak drug identity across splits and inflate metrics, while strict drug-disjoint splitting provides a more realistic test of cold-start generalization to unseen drugs. Under drug-disjoint evaluation, the proposed pathway-enhanced RGCN achieved stronger ranking performance than disjoint baselines and produced better calibrated probability estimates, which is important for risk-aware decision support.

Beyond predictive scores, the retrieval-augmented generation module provides evidence-grounded summaries intended to help users understand why an interaction may occur and what outcomes may be plausible. While such explanations are not mechanistic proofs, grounding outputs in retrieved evidence helps connect predictions to biological context and reduces reliance on unverified free-form generation.

Overall, this work suggests that combining leakage-aware evaluation, heterogeneous graph modeling, and evidence-grounded summarization is a practical direction for building more robust and interpretable DDI screening systems, particularly in settings where new or under-studied drugs must be evaluated under uncertainty.

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